

Stage 2 — Theory

$p(w \mid h) \rightarrow$ underlying language-generating distribution

$\downarrow$

never observed
Let

$$C(\text{"the king"}) = 20$$

and

$$C(\text{"the king", "was"}) = 5$$

in training corpus.

Maximum-likelihood estimation (MLE)

$$\hat{p}(\text{was} \mid \text{the king}) = 0.25$$

$$p \neq \hat{p}$$

$\hookrightarrow$ purely based off observation

$$\hat{p} \rightarrow \text{empirical estimate}$$

Suppose:

1.

$$C(\text{king}) = 500$$

$$\hat{p}(w \mid \text{king}) \Rightarrow 500 \text{ observations}$$

2.

$$C(\text{the king}) = 20$$

$$\hat{p}(w \mid \text{the king}) \Rightarrow 20 \text{ observations}$$

$\vdots$

Ultimately:

$$C(\text{of the king}) = 1$$

$$\hat{p}(\text{death} \mid \text{of the king}) = 1$$

$\therefore$ only one observation

For $w' \neq \text{death}$:

$$\hat{p}(w' \mid \text{of the king}) = 0$$

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Add- α smoothing

$$q_{\alpha}(w \mid h) = \frac{C(h, w) + \alpha}{C(h) + \alpha|V|}$$

- p = unknown truth - $\hat{p}$ = raw empirical estimate - q = actual predictive model
For unsmoothed MLE:

$$q = \hat{p}$$

With smoothing:

$$q \neq \hat{p}$$

Why smoothing?

Suppose:

$$C(h, w) = 0$$

with $C(h) > 0$.

Then, for unsmoothed MLE:

$$\hat{p}(w \mid h) = 0$$

Log probability:

$$\log 0 = -\infty$$

Smoothing changes how the model generalises beyond observed counts.

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Notation

1. n -gram with model order n

$$P(w_t \mid w_{t-n+1}, \dots, w_{t-1})$$

2. History length

$$k = n - 1$$

3. How well does q predict text it didn't train on?

Held-out performance

$$q(w_1 \mid h_1), \dots, q(w_N \mid h_N)$$

Negative average log-probability

$$H_{\text{test}} = -\frac{1}{N} \sum_{i=1}^N \log q(w_i \mid h_i)$$

Units:

$$\text{nats/token}$$

Penalty for bad predictions.

This is the **empirical held-out cross-entropy**.

Lower held-out cross-entropy:

$$\Rightarrow \text{better predictive performance}$$

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Perplexity

$$PP = e^H$$

$$\Rightarrow \text{lower perplexity} = \text{better}$$

If:

$$H_A < H_B$$

then:

$$PP_A < PP_B$$

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Key dependent variable — ΔH_n
 Suppose:

$$H_{n-1} = x_1$$

$$H_n = x_2$$

Then:

$$\Delta H_n = x_2 - x_1$$

If:

$$\Delta H_n < 0$$

more context improves prediction.

If:

$$\Delta H_n > 0$$

more context degrades prediction.

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Examine all length- k histories in the training data.
 Let:

$$N_k$$

be the total number of history occurrences.

Let:

$$f_1^{(k)}$$

be the number of distinct histories appearing exactly once,
 a.k.a. ****singletons****.

Then the singleton-history rate is:

$$S_k = \frac{f_1^{(k)}}{N_k}$$